



### Alexa based microwave

Amazon reportedly planning its own Alexa based microwave and some seven other alexa based smart devices. <https://dgit.in/oct18-23-1>

a long time to come, electronic microphones, amplifiers, and electromechanical recorders seeped into almost all major US record labels by the 1950s. This widened the recordable range to a much wider 60 Hz to 6000 Hz and allowed singers to explore higher frequencies. It also gave rise to a very important profession - the sound engineer. A sound engineer could now capture sound in incredible detail and balance using multiple microphones connected to multi-channel amplifiers. The availability of amplifiers also allowed previously weaker instruments like the guitar to gain equal prominence in music as compared to more powerful instruments like wind and horn instruments.

Another major advancement of this era was sound-on-film technology. Audio waves to be recorded were used to modulate a light source that would be photographed to record the width and density of a "sound track". This also gave rise to the ability to 'multi-track' where multiple separate recordings are played together to generate a composite sound.

### MAGNETIC TAPES

After electrical recordings, perhaps the biggest transformation came about to audio recording through magnetic tapes. Wire recording technology had been around for a while but it was magnetic tapes that made the introduction of magnetism into recording technology practical, and improved the signal to noise ratio greatly. This is the technology that went into cassette tapes that were all the rage up until the advent of CDs. A tape consists of an oxide side that can be magnetically manipulated using a tape head. The substrate on the other side is simply used for strength. An adhesive binds



A magnetic cassette tape

the two sides together. Invented around 1928 in Germany, magnetic tapes didn't reach the rest of the world until the end of World War II. The tape head, initially needle shaped, was later changed to a ring shaped design around 1933 to avoid shredding. The technique of AC biasing also improved sound fidelity for magnetic tapes.

A magnetic tape cassette works on the principle of electromagnetism. The tape head has a coil wound around it with gaps. A current passing through this coil aligns the magnetic material on the tape according to the sound that needs to be recorded. Playback comprises of the opposite action, where the magnetic material influences the signal flowing through the coil and these fluctuations are used to generate the sound.



A digital audio workstation is the modern standard for professional audio recording

### DIGITAL ERA

The first popular digitally recorded music album was released in 1979. Digital audio technology was the first advancement that did not record a complete analogue of the audio to be recorded. Instead, it recorded a stream of digital samples. Quite soon, Sony along with Philips, introduced the Compact Disc. A vast improvement in size and storage capacity, the compact disc replaced almost all pre-existing recording mediums. However, even its dominance was short lived. With the advent of highly efficient compression algorithms and compact audio formats like mp3 and wav, an entirely new category of devices was born, one that could store immense quantity of audio, fit into your pocket and not cost you a bomb - the portable music player that we know today. 📱



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